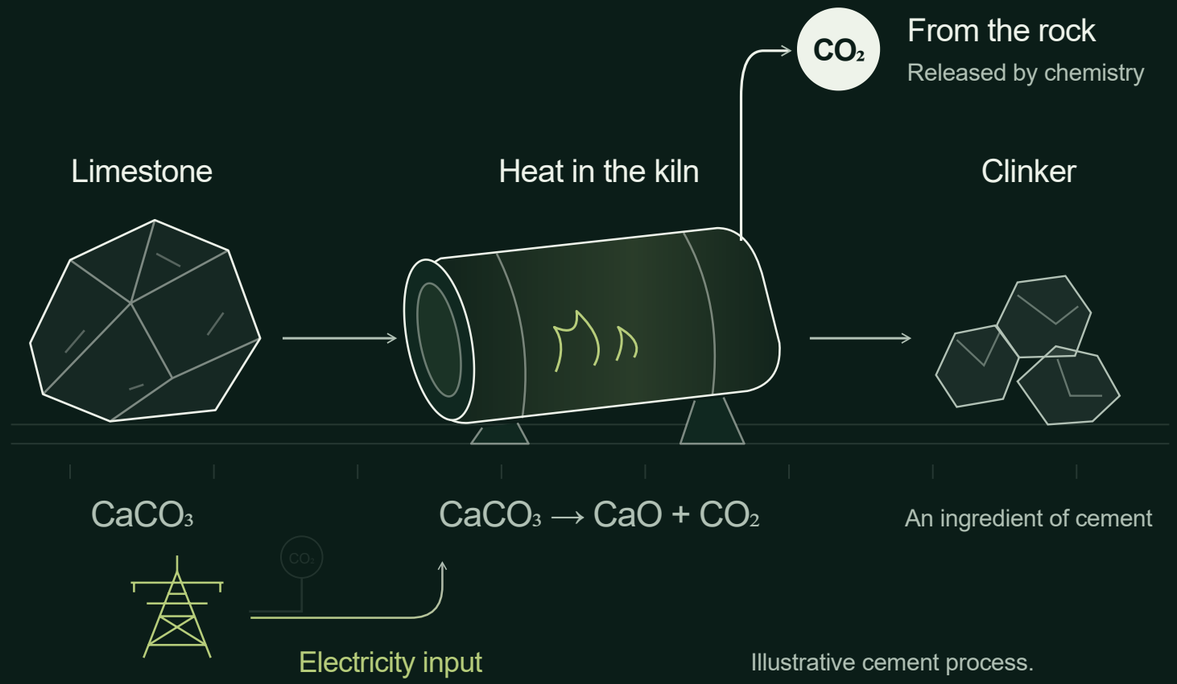
Cleaner energy. A material problem remains.

We are changing how buildings are powered. We also have to change what they are made from.

≈37% of global CO_2 emissions come from buildings and construction.



Illustrative cement process.
Not an emissions model.

Cleaner electricity cannot remove the CO_2 released by limestone chemistry.

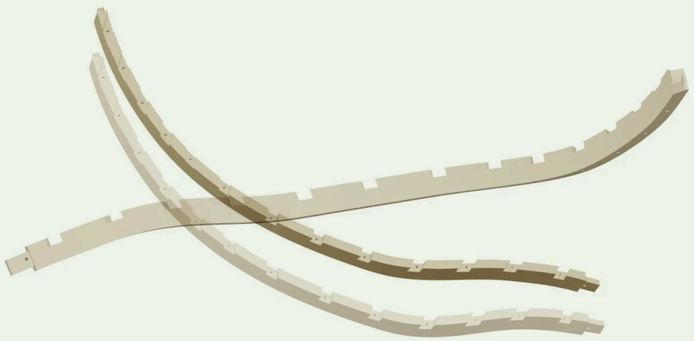
Sources: UNEP 2025–2026 · IEA, cement chemistry

Other emissions and abatement routes omitted.

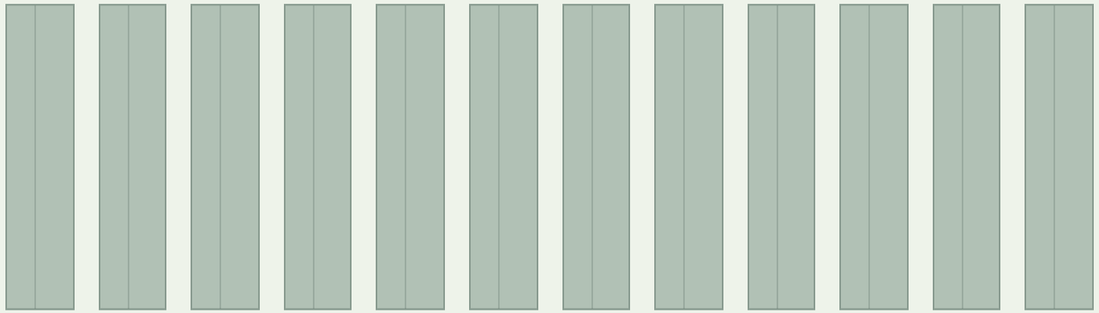
Nature varies.

Making must be repeatable.

Biological materials offer another way to build. Their forms, properties and growth demand a process that can adapt.



CW-001 CW-020 CW-040



REPEATABLE STOCK / SCHEMATIC

Variability is the essence of nature and the enemy of industry. Bower's bet is to make variability manufacturable.

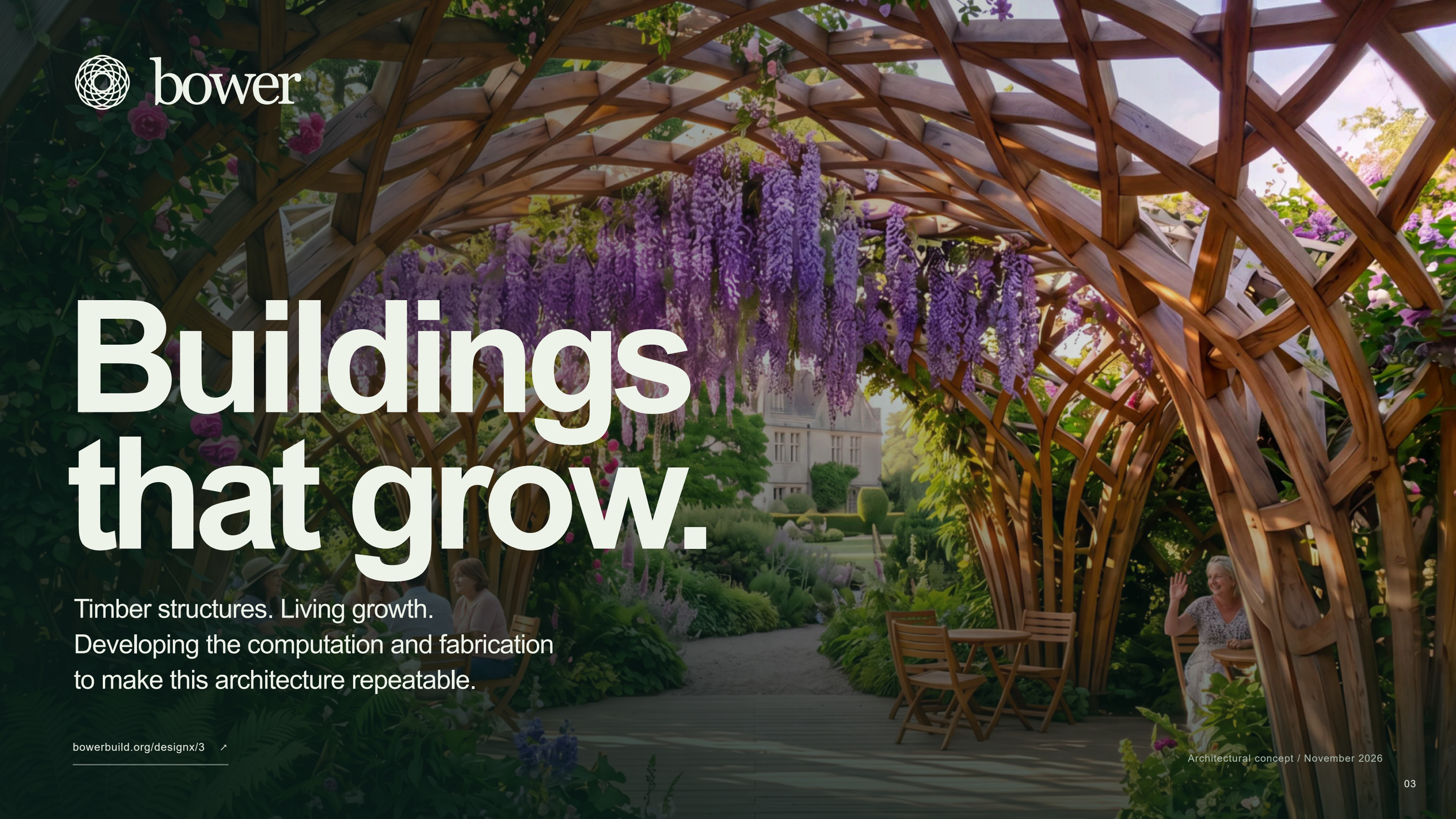
Bower One R13 digital components / standard-stock schematic. Biological materials: UNEP, Building Materials and the Climate ↗

bower

Buildings that grow.

Timber structures. Living growth.
Developing the computation and fabrication
to make this architecture repeatable.

bowerbuild.org/designx/3 ↗



Architectural concept / November 2026

More alive with time.

To add: Clay’s plant-growth simulation.



01 **Structure**

A room from the beginning.



02 **Establishment**

The plants find their place.



03 **Maturity**

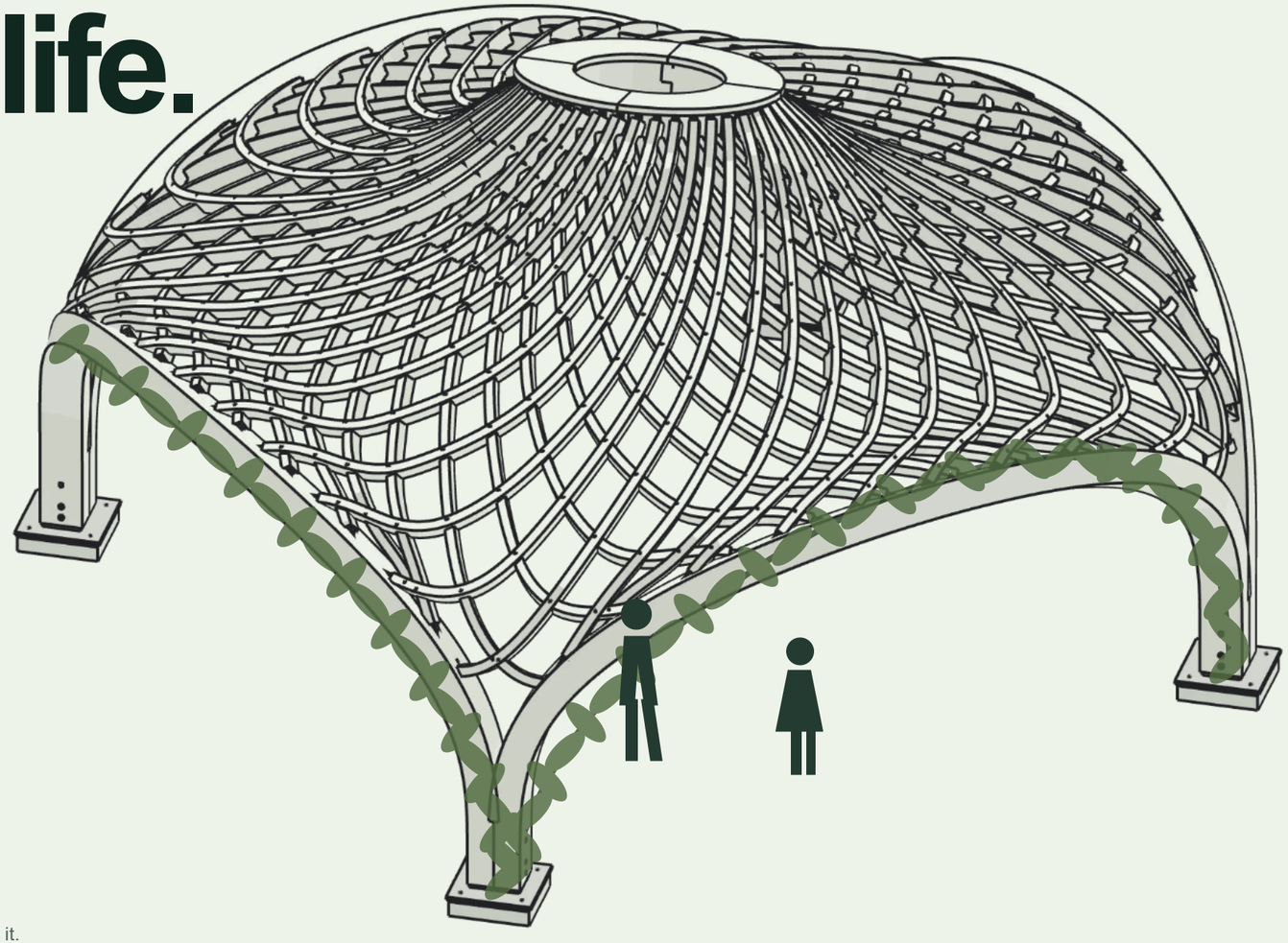
The garden transforms the room.

Illustrative growth states

A timber structure. Designed for life.

Bower assembly

To add: Timber members assemble into the Bower. Plants then colonise the frame.



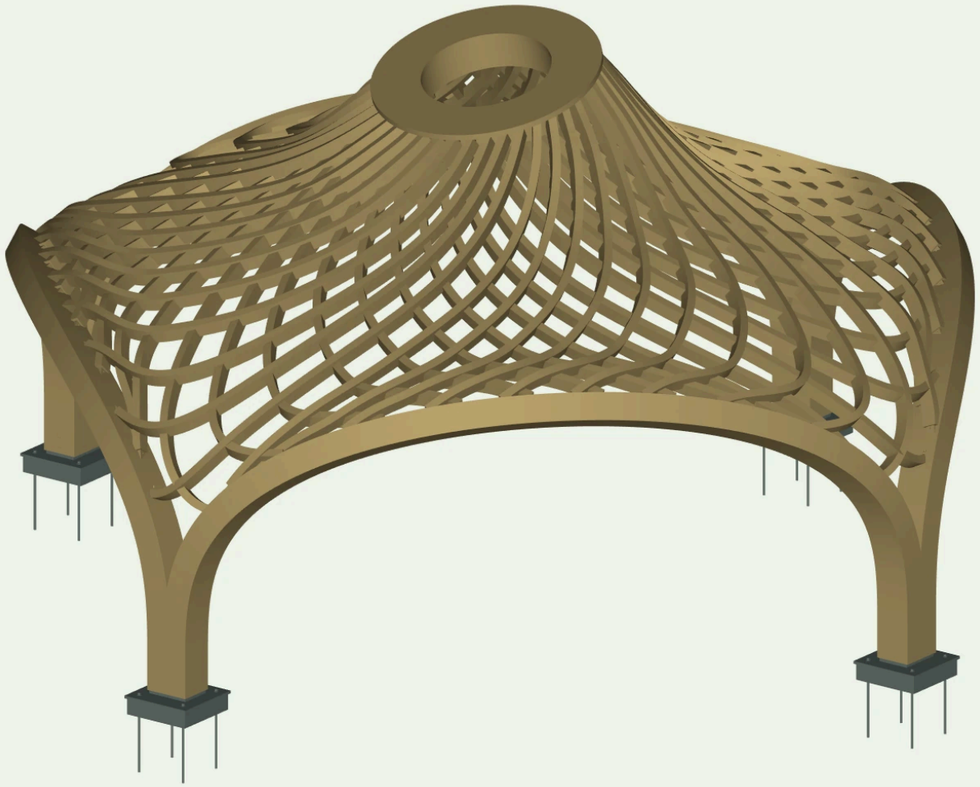
Digital assembly study. The engineered timber frame carries the structure; planting colonises it.

From form to fabrication.

Turn a changing form into parts, checks and fabrication instructions.

The engine diagram

To add: Daniel’s geometry, checks and fabrication workflow. Show how they connect.



Bower One R13 · 939 model parts · 92 primary timbers

Design-to-manufacture engine in development. Existing R13 model shown; current engine diagram to follow.

Different parts.
A repeatable process.

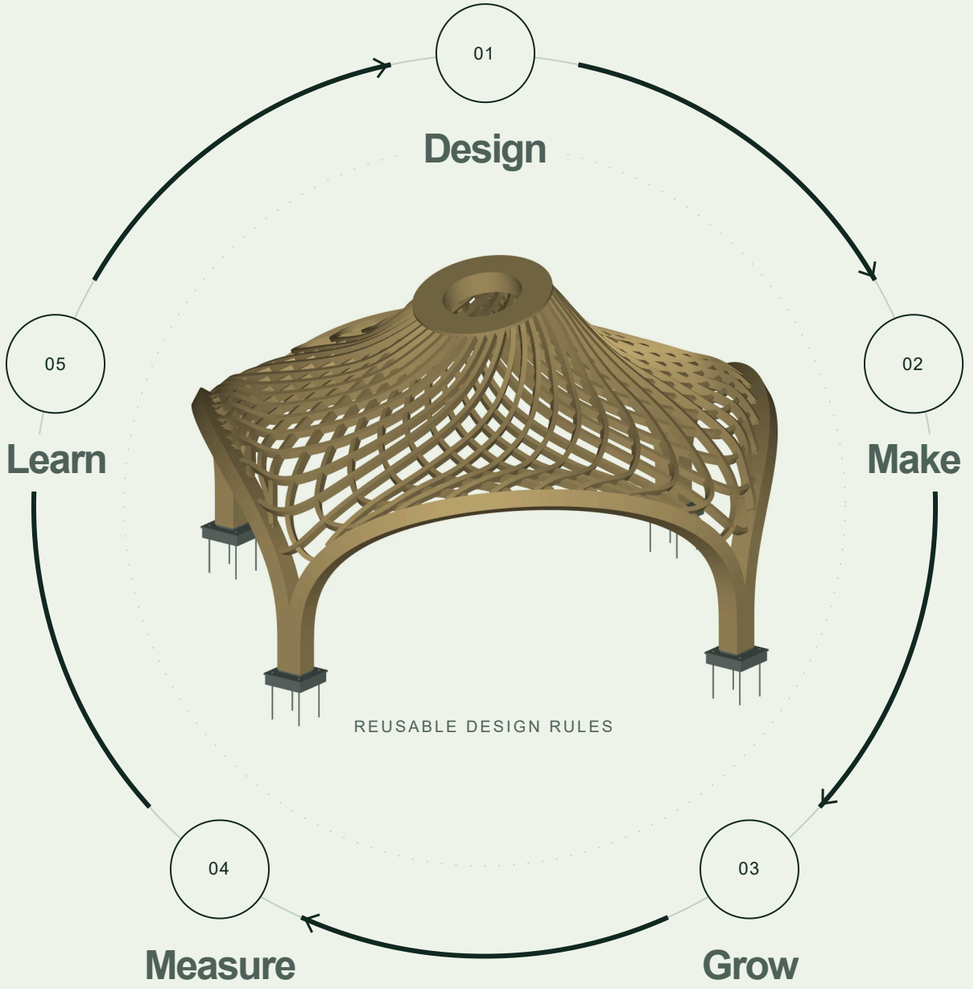
The aim: repeat the method, even when the components change.

Factory floor drawing

To add: Daniel’s live factory plan. Reveal the route from timber to finished component.

Factory design in development. Drawing to follow.

Turn each build into better design rules.



↻ THE NEXT CYCLE

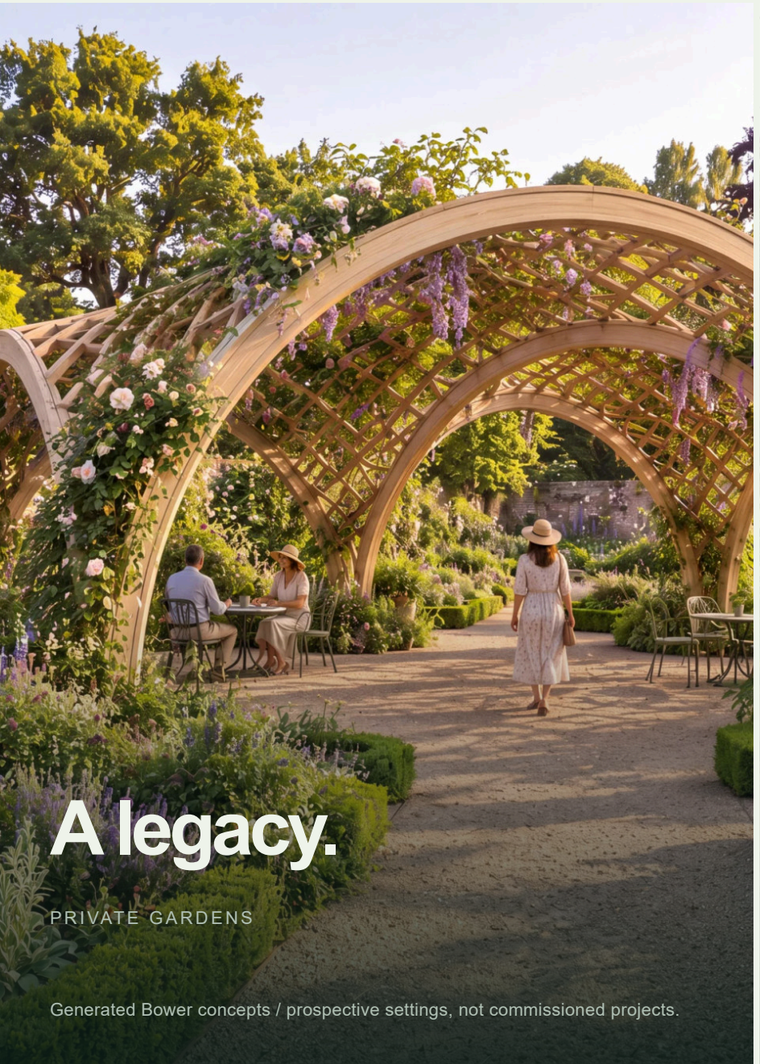
A better starting point. Every time.

What the last Bower teaches the next.

Design → Make → Grow → Measure → Learn → better design.

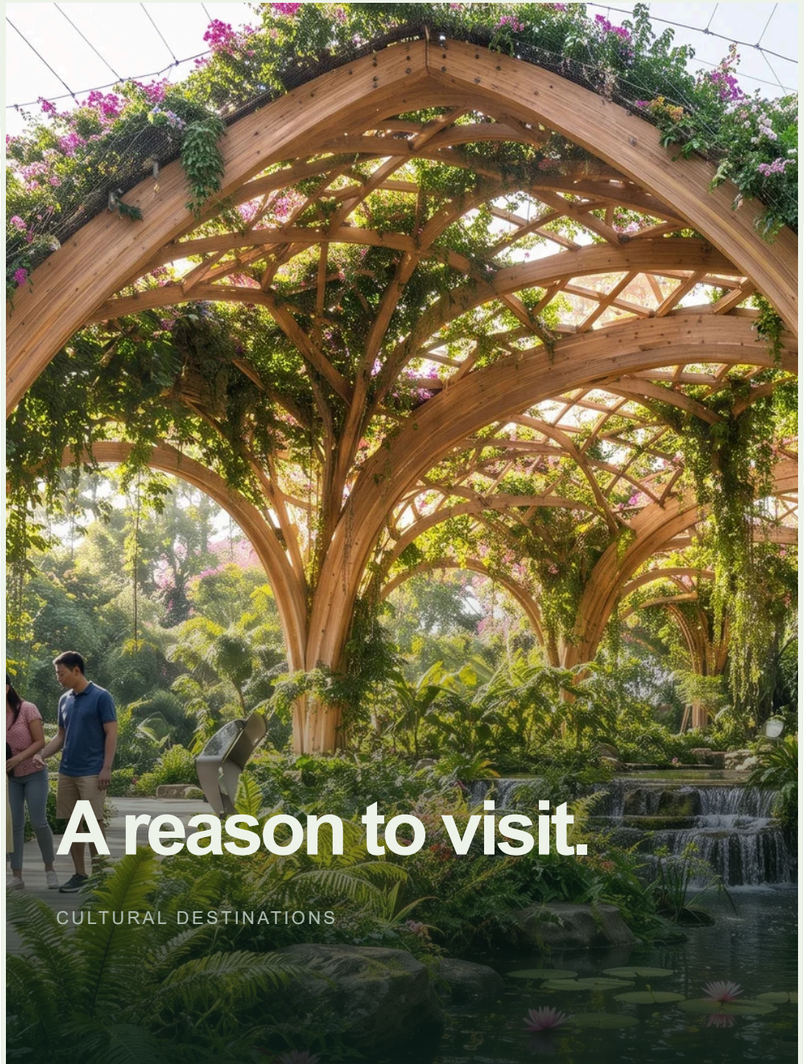
Intended learning system: fabrication and growth data would update design rules. Factory and garden measurement are not yet deployed.

The first customers.



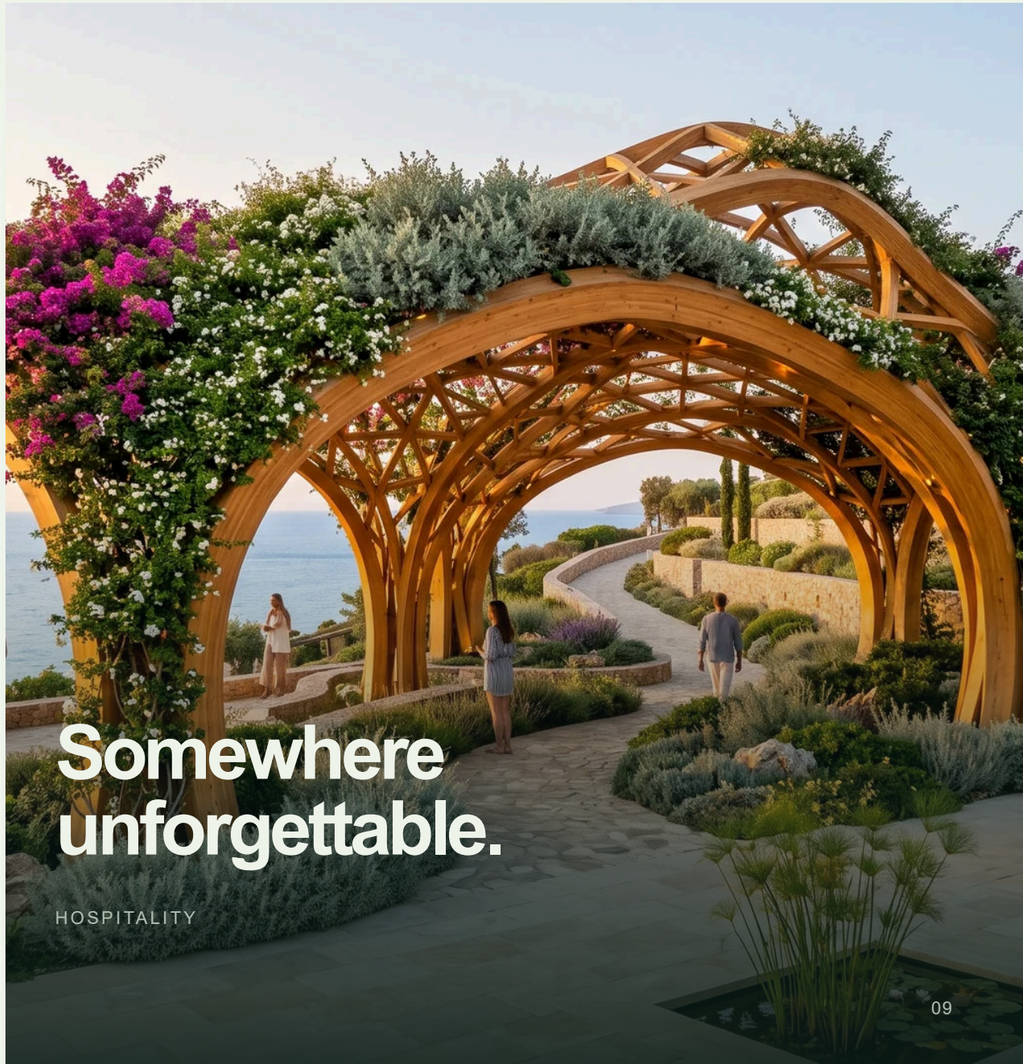
A legacy.

PRIVATE GARDENS



A reason to visit.

CULTURAL DESTINATIONS



Somewhere unforgettable.

HOSPITALITY

Generated Bower concepts / prospective settings, not commissioned projects.

In discussion with

Eden Project

The Newt

25hours



Active discussions, not signed commissions. / Bower concept pictured.

Sell a Bower.
Develop the system.

Each early commission should create a place people want—and knowledge the next build can reuse.

Siting study

£65k

indicative

→

Design development

£100–250k

proposal-specific

→

Commissioned Bower



Build terms by contract.

↓

Reusable design rules

Tested fabrication workflows

Production tooling

Intended commercial model / indicative proposal fees. Capability development and reinvestment depend on project economics and additional finance.

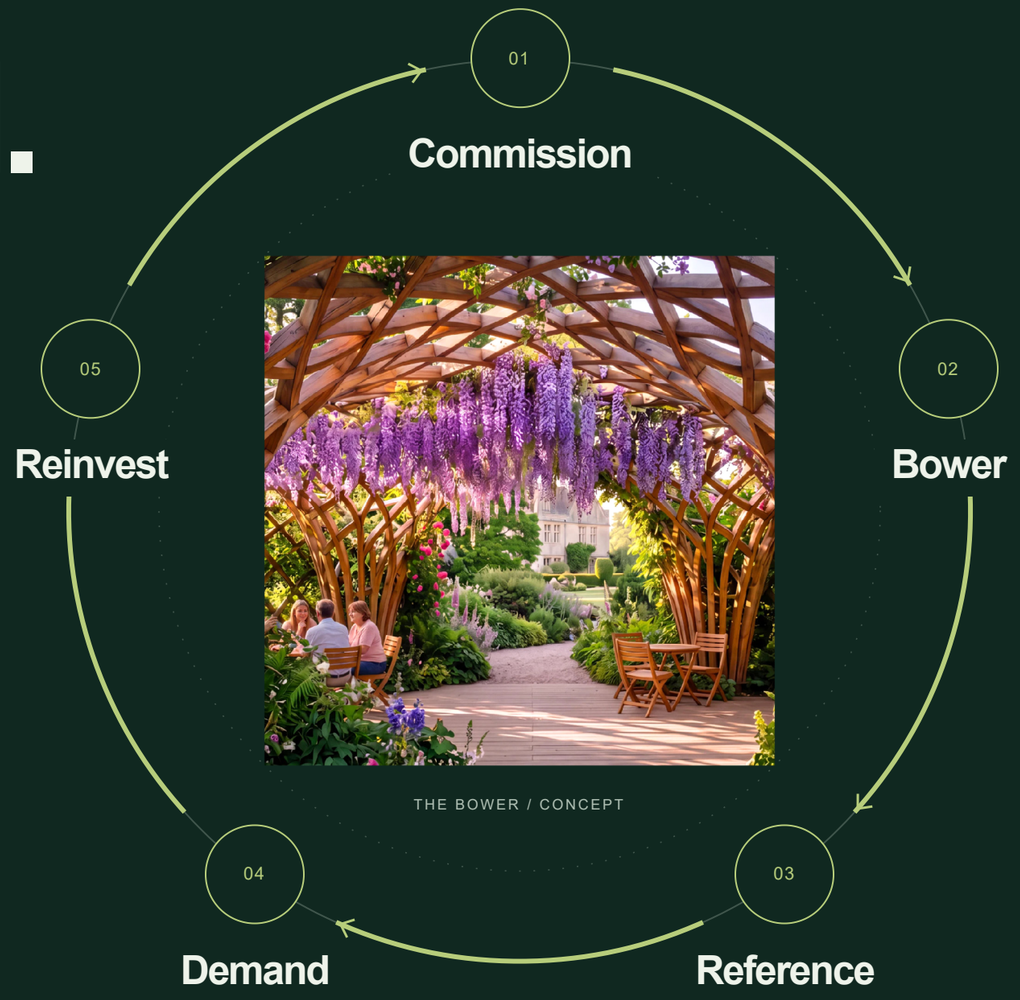
Built proof creates demand.

↻ THE NEXT CYCLE

The next place starts with more behind it.

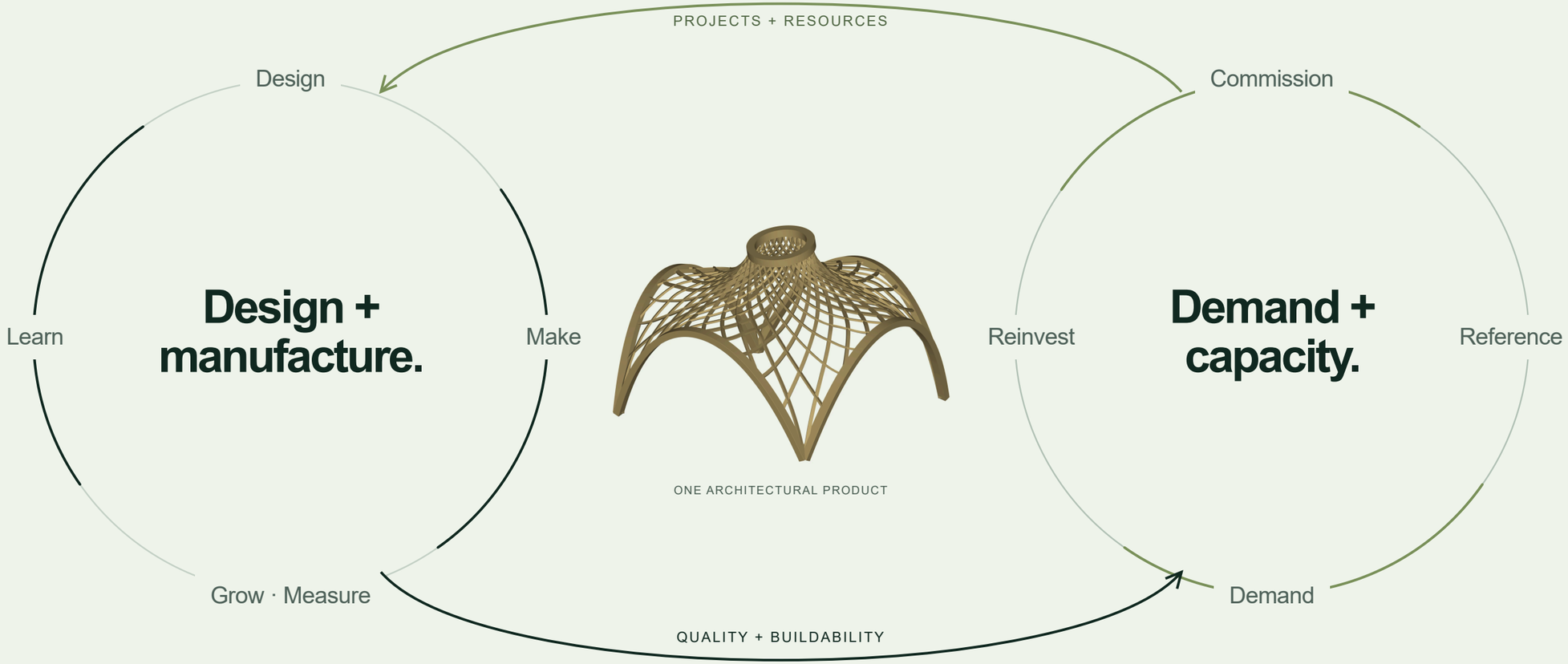
Experience, references and stronger capability.

Commission → Bower → Reference → Demand → Reinvestment.



Intended commercial model. Reinvestment depends on project economics and finance.

From commissions to production capability.



Projects support learning. Learning improves projects.

The long-term objective is repeatable production. First commissions test the product, process and demand; the factory remains in development.

Architecture.
Software. Manufacturing.



Clay Seifert

Architecture.
Computation.
Living systems.

UT Austin / B.Arch



Daniel Guerra

Software.
Fabrication.
Project delivery.

Forsite Studio / project experience

A room is the beginning.

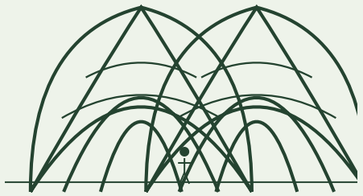
Prove the room. Develop the system. Explore what it could make possible.

01



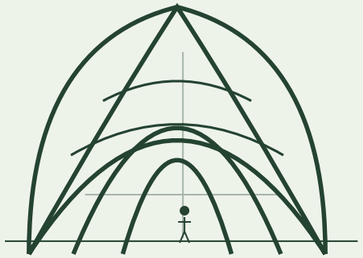
Garden room

02



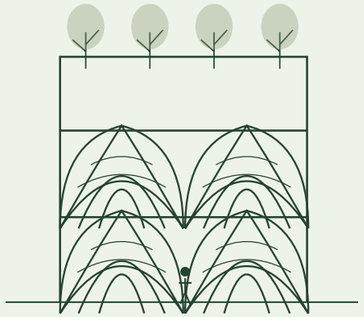
Cultural pavilion

03



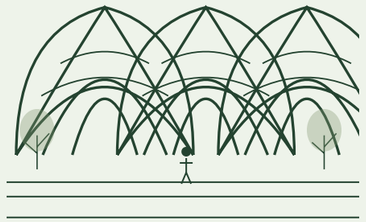
Sanctuary

04



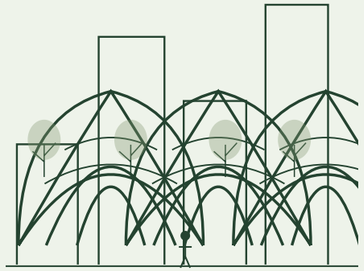
Living building

05



Planted street

06



Urban fabric

FIRST PRODUCT

LONG-TERM RESEARCH HORIZON ↗

Long-term ambition. Speculative scale studies; engineering feasibility and lifecycle carbon performance remain to be demonstrated.

bower

Buildings that grow.

Make the first Bower possible.
Develop the means to make many.

bowerbuild.org/designx/3 ↗

Architectural concept